

Krishnaa Vadivel | Machine Learning Engineer | Scientific ML | Equivariant Models

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Profile

Machine-learning-focused PhD researcher building scientific ML workflows for molecular and materials problems. Combines model development, data/compute infrastructure, and theory fluency across equivariant graph networks, diffusion-style generative models, and retrieval-driven automation for scientific software.

Research Experience

Yale University, Electrical Engineering

Research Assistant / PhD Researcher

2021–present

- Develop scientific ML approaches around first-principles materials workflows, with emphasis on equivariant models, molecular-property prediction, and acceleration of computational science pipelines.
- Build distributed compute and data-processing workflows in Python and C++ using MPI, OpenMP, CUDA, ROCm, PETSc, and SLEPc across national HPC environments including Frontier, Frontera, and Perlmutter.
- Connect ML methods to domain theory through work in first-principles exciton calculations, many-body modeling, and numerical simulation, enabling more realistic scientific-ML problem framing.
- Develop structured notes and derivations for diffusion models, transformer-style operations, and index-notation ML formulations to support rigorous understanding and communication.
- Co-authored a 2025 *Journal of the American Chemical Society* paper and delivered APS presentations in 2024, 2025, and 2026 on self-trapped exciton formation and related excited-state problems.

Professional Experience

Probe Technologies Inc. / Weatherford Wireline Services

Software and Embedded Systems Engineer

2019–2021

- Built CUDA-accelerated data-processing and visualization software for large waveform datasets, strengthening experience in GPU-enabled high-throughput pipelines.
- Contributed ASP-style services and Microsoft SQL-backed workflows for engineering data systems and operational tooling.
- Worked close to embedded and instrumentation contexts, improving practical understanding of data generation, system integration, and performance constraints.

Selected Projects

Equivariant Molecular Property Modeling on QM9

- Implemented graph-based molecular-learning workflows using PyTorch, torch_geometric, and equivariant modeling ideas to predict molecular properties on QM9-style data.

Agentic RAG for Scientific Workflows

- Built retrieval-driven tooling that reads software documentation and academic papers to generate structured starter inputs and workflow scaffolds for computational science and quantum-chemistry-style tasks.

Diffusion and Transformer Theory Notes

- Organized derivations and implementation-oriented notes for DDPM-style diffusion models, conditioning, transformer abstractions, and index-notation formulations.

Distributed Scientific ML Infrastructure

- Combined HPC simulation workflows with CUDA/ROCm acceleration and scalable Python/C++ tooling to support ML-adjacent scientific workloads.

Education

Yale University

PhD, Electrical Engineering

2021–present

Ongoing work spanning scientific computing, first-principles materials workflows, and ML-assisted acceleration.

Yale University

MPhil, Electrical Engineering

2023

Yale University

MS, Electrical Engineering

2023

Cornell University

MEng, Electrical and Computer Engineering

2017–2018

Cornell University

BS, Electrical and Computer Engineering

2013–2017

Selected Coursework

Machine Learning: Probabilistic Machine Learning

Theory: Theoretical Computer Science; strong mathematical/theory preparation supporting modern ML work

Scientific Foundations: Graduate Quantum Mechanics; Solid State Physics; Many-Body Quantum Physics

Engineering Foundations: Signals and systems, embedded systems, and numerical/scientific computing practice

Technical Skills

ML: PyTorch, PyTorch Geometric, scikit-learn, equivariant graph neural networks, graph transformers, diffusion-model theory

Languages: Python, C++, CUDA, JavaScript, Bash, PowerShell

Scientific Python: NumPy, pandas, SciPy, SymPy, matplotlib

Compute: MPI, OpenMP, CUDA, ROCm, PETSc, SLEPc, mpi4py, petsc4py, slepc4py

Platforms: Linux, Docker, Docker Compose, Kubernetes, gcloud, Frontier, Frontera, Perlmutter